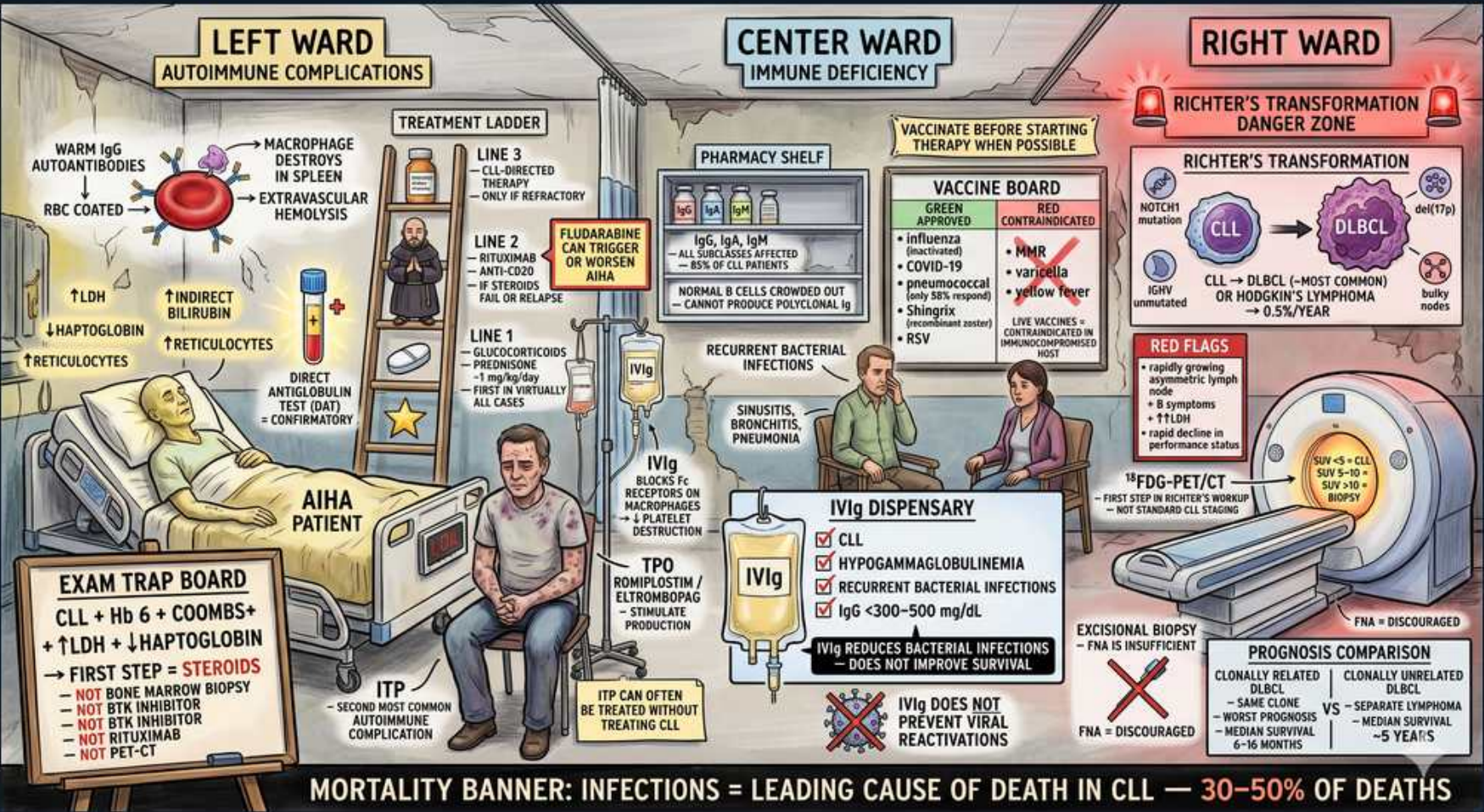


CLL Complications: Autoimmune Cytopenias, Immune Deficiency & Richter's



1. AIHA — warm IgG autoantibody

WHAT YOU SEE

Left ward: RBC coated by warm IgG autoantibodies

WHAT IT ENCODES

CLL-associated autoimmune hemolytic anemia is WARM (IgG-mediated). IgG-coated RBCs are removed extravascularly by splenic macrophages.

BOARD TRAP

CLL-AIHA is WARM IgG, not cold agglutinin — cold (IgM) hemolysis is more typical of lymphoplasmacytic lymphoma/Mycoplasma, not classic CLL.

2. Extravascular hemolysis in spleen

WHAT YOU SEE

Macrophage destroying RBC in the spleen

WHAT IT ENCODES

Splenic macrophages phagocytose IgG-coated RBCs -> extravascular hemolysis: low haptoglobin, high LDH, high indirect bilirubin, reticulocytosis, spherocytes.

BOARD TRAP

Extravascular hemolysis gives spherocytes and splenic destruction; intravascular markers (hemoglobinuria, schistocytes) point elsewhere.

3. Direct antiglobulin (Coombs) positive

WHAT YOU SEE

Direct antiglobulin / Coombs test tube; labs: high indirect bili, high LDH, low haptoglobin, reticulocytosis

WHAT IT ENCODES

Diagnosis of AIHA = positive direct antiglobulin test (DAT/Coombs) plus hemolysis labs (low haptoglobin, high LDH, high indirect bilirubin, reticulocytosis).

BOARD TRAP

A positive DAT distinguishes AIHA from non-immune hemolysis and from disease progression — falling Hgb with positive Coombs is autoimmune, not marrow failure.

4. AIHA treatment ladder (steroids first)

WHAT YOU SEE

Treatment ladder: line 1 steroids, line 2 rituximab +/- chemo

WHAT IT ENCODES

First-line AIHA = corticosteroids. Refractory cases: rituximab, IVIG, splenectomy, or CLL-directed therapy if hemolysis is disease-driven.

BOARD TRAP

First-line for CLL-AIHA is STEROIDS, not rituximab and not starting CLL chemo — autoimmune cytopenia alone treated with steroids.

5. Fludarabine can trigger AIHA

WHAT YOU SEE

Sign: fludarabine can trigger/worsen AIHA

WHAT IT ENCODES

Purine analogs (fludarabine) can precipitate or worsen autoimmune hemolytic anemia in CLL — avoid fludarabine in patients with active AIHA.

BOARD TRAP

Fludarabine is contraindicated with active AIHA because it can severely worsen hemolysis — a specific drug-avoidance board point.

6. Immune-mediated thrombocytopenia

WHAT YOU SEE

Exam trap board (left) plus the immune low-platelet figure as second autoimmune complication

WHAT IT ENCODES

Immune-mediated low platelets is the second autoimmune cytopenia of CLL; can be treated with steroids without treating the underlying CLL if isolated.

BOARD TRAP

Like AIHA, immune thrombocytopenia does NOT upstage to Rai IV — it's autoimmune, treated with steroids, not an automatic indication for CLL chemo.

7. Exam trap: CLL + anemia + Coombs+

WHAT YOU SEE

Exam trap board: CLL + low Hgb + Coombs+, high LDH, low haptoglobin -> first line steroids

WHAT IT ENCODES

Classic vignette: CLL patient with new anemia, positive Coombs, high LDH, low haptoglobin = AIHA. First-line is steroids; not rituximab, not bone marrow biopsy, not PET.

BOARD TRAP

Don't jump to marrow biopsy or PET-CT for the anemia — positive Coombs makes it autoimmune; treat with steroids.

8. Immune deficiency / recurrent infections

WHAT YOU SEE

Center ward: pharmacy shelf, recurrent bacterial infections (sinusitis, bronchitis, pneumonia)

WHAT IT ENCODES

Hypogammaglobulinemia from CLL causes recurrent encapsulated bacterial infections (sinusitis, bronchitis, pneumonia). Infection is the leading cause of death.

BOARD TRAP

The dominant immune defect in CLL is HUMORAL (hypogammaglobulinemia) -> encapsulated bacteria; not primarily a T-cell/opportunistic defect at baseline.

9. Vaccinate before therapy

WHAT YOU SEE

Vaccine board: inactivated vaccines approved, live vaccines contraindicated

WHAT IT ENCODES

Give inactivated vaccines (influenza, pneumococcal, RSV, Shingrix) early, ideally BEFORE starting therapy when response is better. LIVE vaccines are contraindicated.

BOARD TRAP

Live vaccines (e.g. live zoster, MMR) are CONTRAINDICATED in CLL/immunosuppressed; use the inactivated recombinant zoster vaccine instead.

10. IVIG for recurrent infection + low IgG

WHAT YOU SEE

IVIG dispensary: indicated for hypogammaglobulinemia with recurrent infections, IgG <300-500

WHAT IT ENCODES

IVIG is indicated when there is hypogammaglobulinemia (IgG <300-500) WITH recurrent serious bacterial infections — it reduces infections but does NOT improve survival.

BOARD TRAP

IVIG reduces infection frequency but does NOT prevent viral reactivation and does NOT improve overall survival — and is not given for low IgG alone without infections.

11. Richter's transformation to DLBCL

WHAT YOU SEE

Right ward danger zone: CLL -> DLBCL, FDG-PET hot node

WHAT IT ENCODES

Richter's transformation = CLL evolving to aggressive DLBCL (most common) or Hodgkin lymphoma. Suspect with rapid nodal growth, B symptoms, high LDH; FDG-PET hot (SUV>10) node guides EXCISIONAL biopsy.

BOARD TRAP

Richter's is confirmed by EXCISIONAL biopsy of the PET-avid node — FNA is insufficient. NOTCH1/TP53 mutations predispose; prognosis is poor (median ~5 yr if clonally unrelated, much worse if related).

12. Mortality banner: infection

WHAT YOU SEE

Bottom banner: infections = leading cause of death in CLL, 30-50% of deaths

WHAT IT ENCODES

Infections cause 30-50% of CLL deaths — the single leading cause — driven by hypogammaglobulinemia, complement defects, and therapy-related immunosuppression.

BOARD TRAP

Leading cause of CLL death is INFECTION, not bleeding, not Richter's transformation, and not progressive marrow failure.
